



ROBERTO NOGUEIRA

AI-Native Engineer | Ruby on Rails | React | Agentic Systems
Software & Telecom Engineer | AI-Native Product Architect
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PROFESSIONAL SUMMARY

AI-Native Software **Engineer** with a strong Telecom background and **7+ years** of **Ruby on Rails** experience. Skilled in **Ruby, React, JavaScript, Python, and Elixir**, building agentic systems, AI-native knowledge systems, modular monoliths, microservices, scalable backend architectures, CI/CD, automation, and workflow integration. Deep experience in **OSS/BSS**, multinational system integration, **billing, payroll, subscriptions, API integrations**, mentoring, technical leadership, and reliable software delivery at scale.



VALUE HIGHLIGHTS

- ✓ **7+ years** of experience with **Ruby on Rails** and strong backend / full-stack delivery.
- ✓ **International engineering experience** across LATAM, USA, Canada, and Europe.
- ✓ **300%** productivity improvement through Ruby/Rake automation at Ericsson.
- ✓ Experience with **payroll, billing, subscriptions**, event-driven integrations, and async processing.
- ✓ **AI-assisted SDLC, agentic workflows, multi-agent systems**, and context engineering.



PORTFOLIO

REAL PROJECTS. REAL IMPACT.

★ FEATURED PROJECT

HOLOFLUX

A human-centered semantic runtime and visual language for intent, inquiry, meaning, evidence, execution, and evolving understanding.



github.com/enogrob/project-holoflux

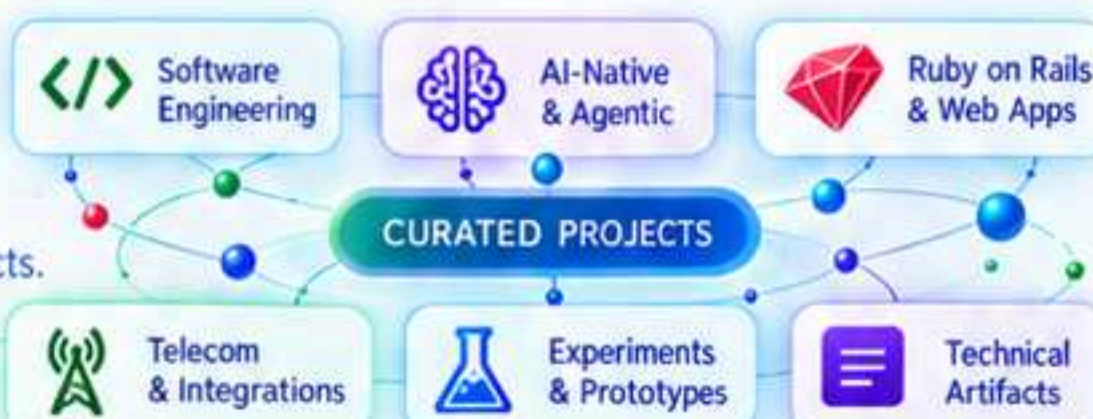


PROJECTS PORTFOLIO

A curated portfolio of software engineering, AI-native, Ruby on Rails, telecom, and product projects spanning platforms, integrations, experiments, and technical artifacts.



enogrob.github.io/projects/



CORE COMPETENCIES

	Languages:	Ruby, Python, JavaScript, Elixir, Erlang/OTP, Java
	Frameworks & Frontend:	Ruby on Rails, React, Phoenix, Vue, jQuery, Bootstrap
	AI & Agentic Engineering:	AI-Native Software Engineering, AI-Assisted SDLC, Agentic Workflows, Multi-Agent Systems, Context Engineering, GitHub Copilot, OpenAI Codex, Claude Code, Warp
	Architecture:	Modular Monoliths, Packwerk, Microservices, REST, GraphQL, Clean Architecture, DDD, Event-Driven Architecture, Kafka
	Backend & Async Processing:	Sidekiq, Redis, Background Jobs, API Integration
	Databases:	PostgreSQL, MySQL, Oracle/PLSQL, MongoDB, Neo4j, Sybase
	Testing & Quality:	RSpec, Minitest, PyTest, JUnit, ExUnit, SimpleCov, RubyCritic, Skunk
	Security:	Brakeman, bundler-audit, bundler-leak
	DevOps & CI/CD:	Git, GitHub, GitHub Actions, GitLab, Bitbucket, Docker, Kubernetes, Terraform, Helm, Puppet
	Cloud & Platforms:	AWS, Heroku, Engine Yard, OpenStack, StarlingX, IBM Cloud
	Observability & Tools:	Datadog, Kibana, Redash, Rollbar, OpenAPI/Swagger, Postman
	Telecom & OSS/BSS:	Ericsson OSS-RC, BSS, MSS, APG, AXE, CPP, MOShell
	Engineering Practices:	Clean Code, SOLID, DRY, Design Patterns, Scrum

TURNING COMPLEXITY INTO UNDERSTANDING



PROFESSIONAL EXPERIENCE

1**06/2013 — Present****Zoatworks** — São José dos Campos, Brazil**Founder & Consultant | AI-Native Software Engineer**

★ **Summary:** Providing software engineering and technology consulting focused on AI-native development, Ruby on Rails, systems integration, telecom, and backend platforms.

★ **Key achievements:**

- Built multi-agent and AI-first frameworks for eBooks, career workflows, technical blogs, and automation.
- Built an event-driven Ruby on Rails microservice with Kafka to orchestrate student subscription workflows and external API integrations.
- Integrated AI-driven workflows into GitHub-based engineering processes.
- Used AI and code-quality tooling to automate Rails technical-debt analysis and modernization prioritization.
- Developed scalable backend services with Ruby, Elixir, and Erlang/OTP.



Projects: AI-native frameworks — COSMOS / HOLOFLUX / IOP / FACTORIES | Anubis/CX (Qeevo) | PicPay (IEBT) | Track Server (ITER) | Adaive (Lab360) | Taxweb Auditor (GSW) | OSS Support (Entel)

2**06/2021 — 09/2024****Encora** — Campinas, Brazil**Senior Ruby on Rails Engineer | System Specialist**

★ **Summary:** Worked on payroll, billing, and Employer Risk systems at Justworks, plus cloud/edge initiatives at Wind River.

★ **Key achievements:**

- Developed scalable payroll and billing workflows using Ruby on Rails, React, GraphQL, Redis, and Sidekiq.
- Worked on Workers' Comp billing including rates, policies, eligibility, error handling, fallback, and reconciliation.
- Built reliable asynchronous processing solutions with Sidekiq/Redis, including bulk S3 workflows.
- Maintained a Rackwerk modular monolith and modernized legacy interfaces with React.
- Used CI/CD and observability with GitHub Actions, Datadog, Kibana, Redash, and Rollbar.
- Technical leadership: interviewing, mentoring, and collaboration.



Projects: PEO / Payroll Platform | Time Tracker | WRCP (Wind River)

3**07/2018 — 06/2021****InMov** — São Paulo, Brazil**Senior Ruby on Rails Engineer (Full Stack)**

★ **Key achievements:**

- Developed Smart City applications using Ruby on Rails, JavaScript, jQuery, and AJAX.
- Led Ruby on Rails upgrades from version 4.2 to 6.0, improving performance, security, and maintainability.
- Created Site Rails Manager, a Bash/Docker CLI toolkit to automate and standardize Ruby on Rails development environments and workflows.



Projects: City Hall Permit System | Site Rails Manager

4**12/2004 — 06/2013****Ericsson** — Indaiatuba, SP, Brazil**Software Developer | Solution Integrator | Project Manager**

★ **Key achievements:**

- Developed, integrated, and deployed telecom systems in large-scale OSS/BSS environments.
- Led international OSS/BSS initiatives across LATAM, USA, Canada, and Europe.
- Built MOP Automation using Ruby and Rake, achieving 300% productivity improvement.
- Delivered technical training and knowledge transfer in international projects.



Projects: BSS Dual Stack | Full IP Rehoming | MOP Automation | Revenue Assurance (Vivo) | OSS/BSS Upgrades & Support



EDUCATION

1998 — Data Communications — USP

1994 — Software Engineering (Postgraduate) — UNIAP

1985 — Telecommunications / Electronics Engineering (Bachelor's Degree) — INATEL



CERTIFICATIONS

SAP Integration — Thomson Reuters

Solution Integrator Experienced (Job Stage 5) — Ericsson

Getting and Cleaning Data — Johns Hopkins University

Functional Programming in Erlang — Kent University

R Programming — Johns Hopkins University



COURSES & SPECIALIZATIONS

2026 — Claude Code: The Complete Course — Udemy

2026 — GitHub Copilot CLI — GitHub

2025 — AI Automation — Udemy

2025 — MCP Crash Course — Udemy

2025 — Working with Vue 3 and Go — Udemy

2025 — Working with Microservices in Go — Udemy

2025 — GitHub Copilot — Udemy

2023 — Metaverse — Coursera

2019 — Elixir & OTP — The Pragmatic Studio

2019 — Elixir: Scalable and Efficient Application Development — Udemy

2017 — Specialization in Erlang Programming — Kent University / FutureLearn

2016 — Specialization in Ruby on Rails Web Development — Johns Hopkins University / Coursera

2015 — Web Applications Architectures — University of New Mexico / Coursera

2014 — Specialization in Data Science — Johns Hopkins University / Coursera

2010 — Ruby on Rails Advanced — Caelum

2010 — Ruby on Rails — Caelum



LANGUAGES

Portuguese — Native / Bilingual

English — Full Professional

Spanish — Intermediate